Exercise # 13.8

3–4: Complete the squares and locate all absolute maxima and minima, if any, by inspection. Then check your answers using calculus.

3: $f(x,y) = 13 - 6x + x^2 + 4y + y^2$

$f(x,y) = (x-3)^2 + (y+2)^2$; minimum at $(3,-2)$, no maxima.

4: $f(x,y) = 1 - 2x - x^2 + 4y - 2y^2$.

$f(x,y) = -(x+1)^2 - 2(y-1)^2 + 4$

maximum $(-1, 1)$, no minima.

10–14: Locate all relative maxima, relative minima, and saddle points, if any.

10: $f(x,y) = x^2 + xy - 2y - 2x + 1$

$f(x,y)$ $f_x = 2x + y - 2 = 0$,

$f_y = x - 2 = 0$; critical points

$(2,-2)$; $D = -1 < 0$ at $(2,-2)$, saddlepoint

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11: $f(x,y) = x^2+xy+y^2-3x$

$f_x = 2x+y-3 = 0$

$f_y = x+2y = 0$; critical point $(2,-1)$;

$D = 3>0$ and $f_{xx} = 2>0$ at $(2,-1)$

relative minimum

12: $f(x,y) = xy - x^3 - y^2$

In final f has a saddle point at $(0,0)$

in final f has a relative maximum at $\left(\frac{1}{6}, \frac{1}{12}\right)$

13: $f(x,y) = x^2+y^2+\frac{2}{xy}$

Relative minimum at the points $(1,1)$ and $(-1,-1)$

14: $f(x,y) = xe^y$

$e^y = f_x(x,y)$

$x \cdot e^y = f_y(x,y)$

At critical points, $e^y = 0$, and $x \cdot e^y = 0$

However, e^y cannot be equal to 0,

Thus, there is no critical points,

and there will be no local extrema

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or saddle points.

12: Solution:

We given that

$$xy - x^3 - y^2 = f(x, y)$$

$$f_x(x,y) = x - 2y = y - 3x^2, \ f_y(x,y)$$

Consider $3x^2 - 3y = 0$ and $3y^2 - 3 = 0$ we get two critical points at $(0,0)$ and $\left(\frac{1}{6}, \frac{1}{12}\right)$

By the theoram 13.8.6, let f be a function of two variables with continuous second-order partial derivatives in some disk centered at a critical point (x_0, y_0) and let,

$$f_{xx}(x_0, y_0)\, f_{yy}(x_0, y_0) - f_{xy}^2 (x_0 - y_0)$$

$$= D$$

Do partial derivatives and check for a critical points at $(0,0)$

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$f_{xx}(x,y) = -6x$, $f_{xy}(x,y) = 1$

and $f_{yy}(x,y) = -2$

$f_{xx}(0,0) = 0$, $f_{xy}(0,0) = 1$ and

$f_{yy}(0,0) = -2$

$D = 0 \times (-2) - (1)^2$

$D = -1 < 0$

Here $D < 0$, so f has a saddle point at $(0,0)$

Check for a critical point at $\left(\frac{1}{6}, \frac{1}{12}\right)$

$f_{xx}(x,y) = -6x$, $f_{xy}(x,y) = 1$

and $f_{yy}(x,y) = -2$

$f_{xx}\left(\frac{1}{6}, \frac{1}{12}\right) = -1$, $f_{xy} = \left(\frac{1}{6}, \frac{1}{12}\right) = 1$

and $f_{yy}\left(\frac{1}{6}, \frac{1}{12}\right) = -2$

$$(-1) \times (-2) - (1)^2 = D$$

$$D = 1 > 0$$

Because here $D > 0$ and ~~f_{xy}~~ $f_{xx}\left(\frac{1}{6}, \frac{1}{12}\right) < 0$, f has a relative maximum at $\left(\frac{1}{6}, \frac{1}{12}\right)$

13: Solution

Step 1: First partial derivatives given the function:

$$f(x, y) = x^2 + y^2 + xy^2$$

Find the first partial derivatives of $f(x, y)$:

$$f_x(x, y) = \frac{\partial}{\partial x}\left[x^2 + y^2 + xy^2\right] = 2x - x^3y^2$$

$$f_y(x, y) = \frac{\partial}{\partial y}\left[x^2 + y^2 + xy^2\right] = 2y - xy^2$$

Step 2: Second partial derivatives Find the 2nd partial derivatives of $f(x, y)$

and $f_x(x,y)$:

$$f_{xx}(x,y) = \frac{\partial^2}{\partial x^2}\left[2x - x^2y^2\right] = 2 - 2x^3y^2$$

$$f_{yy}(x,y) = \frac{\partial^2}{\partial y^2}\left[2y - xy^2\right] = 2 - 2x^2y$$

$$f_{xy}(x,y) = \frac{\partial^2}{\partial y \partial x}\left[2x - x^2y^2\right] = 2xy^2 = 2xy$$

Critical points to find the critical points, we set both first P.D equal to Zero.

For $fx(x,y)$:

$$2x - x^2y^2 = 0$$

For $fy(x,y)$:

$$2y - xy^2 = 0$$

Solving these equations, we find two critical points 1. $x=1$, $y=12$. $x=-1$, $y=-1$ 2nd P.D test using the hessian determinant $D(x,y)$.

$$D(x,y) = [f_{xx}(x,y) \cdot f_{yy}(x,y)] - [f_{xy}(x,y)]^2$$

For the critical point $(1,1)$:

$$D(1,1)$$
$$= [(2-2(1)^3(1)^2 \cdot (2-2(1)^2(1))]$$
$$-[(2(1)(1)^2-2(1)(1)]^2 = 32 > 0$$

For critical points $(-1,-1)$

$$D(-1,-1)$$
$$= [(2-2(-1)^3(-1)^2 \cdot (2-2(-1)^2(-1))]$$
$$-[(2(-1)(-1)^2-2(-1)(-1)]^2 = 32 > 0$$

Since D is positive for both critical points and fun is positive, we have relative minima, at the points $(1,1)$ and $(-1,-1)$ so, the results R.m at $(1,1)$ and $(-1,-1)$